



**Waiting for everyone
to join**

**TU257
Information Systems
Marisa Llorens**

Relational Database principles

Information Systems

Lesson 2 - 25/09/24

TU257

Lesson Outline

- Design Process
- Conceptual Model - Entity Relation model
 - Entities
 - Attributes
 - Relationships

Data Models

- Specify concepts that are used to describe the database. These can be categorised as:
 - ***High Level (Conceptual):***
 - Concepts at user level: entities/classes, attributes, relationships/associations
 - ***Representational (Logical):***
 - Relational, network, hierarchical – may be record based, object-oriented
 - ***Low Level (Physical):***
 - Details of physical data storage; formats, access paths, ordering

Data Models

- Specify concepts that are used to describe the database. These can be categorised as:
 - ***High Level (Conceptual):***
 - Rough ER Diagram + Key Based ER Diagram
 - ***Representational (Logical):***
 - Normalised model
 - ***Low Level (Physical):***
 - Implemented model

Database Design

- Database design deals with how to design a database.
- Importance of good design:
 - Poor design results in unwanted ***data redundancy***.
 - Poor design generates ***errors*** leading to bad decisions.
- Practical approach:
 - Focus on principles and concepts of database design.
 - Importance of logical design.

Major Steps in Database Design

1. Requirements Analysis:

- Talk to the potential users.
- Understand what data is to be stored, and what operations and requirements are desired.

2. Conceptual Database Design:

- Develop a high-level description of the data and constraints

Major Steps in Database Design

3. Logical Database Design:

- Convert the conceptual model to a schema in the chosen data model of the DBMS.
- For a relational database, this means converting the conceptual to a relational schema (logical schema).

Major Steps in Database Design

4. Schema Refinement:

- Look for potential problems in the original choice of schema and try to redesign.

5. Physical Database Design:

- Direct the DBMS into choice of underlying data layout (e.g., index/clustering) in the hope of optimising the performance.

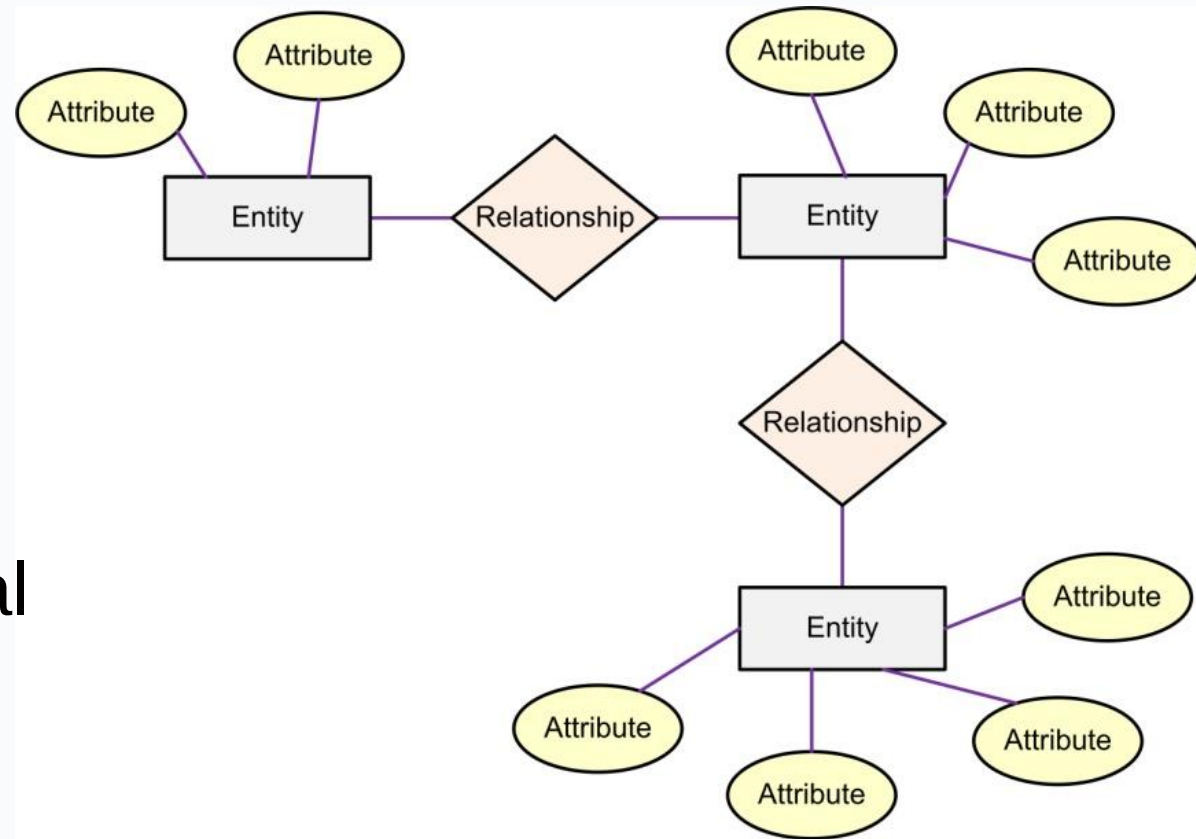
Major Steps in Database Design

6. Applications and Security Design:

- How will the underlying database interact with surrounding applications.

Conceptual Database Design

- **Goal:** Specification of database schema
- **Methodology:**
 - Use E-R model to get a high-level graphical view of essential components of enterprise and how they are related



Entity-Relationship Model

- Entity-Relationship (ER) Model is based on the notion of real-world entities and relationships among them. While formulating real-world scenario into the database model, the ER Model creates entity set, relationship set, general attributes and constraints.

Entity-Relationship Model

- ER Model is best used for the conceptual design of a database.
- ER Model is based on:
 - **Entities** and their *attributes*.
 - **Relationships** among entities.

Entity-Relationship Model: Entities

- **Entity** – An entity in an ER Model is a real-world entity having properties called **attributes**. These are used to describe a particular entity (e.g. name, RSI#, height).
- An **entity** corresponds to a **person**, **place**, **thing**, **event**, or **concept** about which you are interested in recording data i.e.
 - **Person:** EMPLOYEE, STUDENT, PATIENT, **Place:** STORE, WAREHOUSE, COUNTY, **Object:** MACHINE, BUILDING, CAR, **Event:** SALE, REGISTRATION, RENEWAL, **Concept:** ACCOUNT, COURSE, WORK CENTRE

E-R Diagrams: Entities

- An ***entity*** is any object in the system that you want to model and store information about.
 - Individual objects are called ***entities***.
 - Groups of same type objects are ***entity types*** or ***entity sets***.
 - Entities are represented by rectangles (either with round or square corners)

E-R Diagrams: Entities

- Entities are ***equivalent*** to ***database tables*** in a relational database, with each row of the table representing an ***instance*** of that entity.
- Entities are ***abstract concepts***, each representing one or more instances of the concept in question.
- Each entity is represented by a box within the E-R Diagram.

E-R Diagrams: Entities

Type vs. Instance

- **Entity Type** (Entity Class):
 - Entities can usefully be classified into entity types.
 - Collection of similar entities, about which we want to store similar information.
- **Instance:**
 - An entity is an instance of an entity type, e.g. Joe Bloggs may be an instance of the employee entity type.
- You model entity types, not individual instances.
- However, the term '*entity*' is often used to refer to both entity type and instance.

Entity-Relationship Model: Attributes

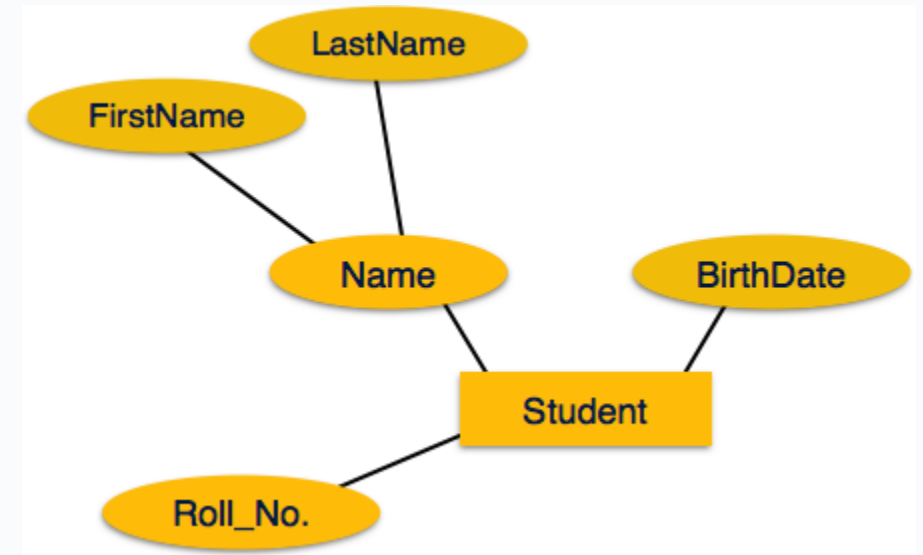
- All the data relating to an entity is held in its ***attributes*** i.e. an attribute is a property of an entity.
- Every **attribute** is defined by its set of values called **domain**
 - name may be a 20 character string;
 - PPS# is seven-digits plus a letter;
 - Student ID may be a char with a letter followed by 8 digits;
 - Price may be a float number with 2 decimal places.

E-R Diagrams: Attributes

- Attributes can be...
 - *simple* or *composite*
 - *single-valued* or *multi-valued*

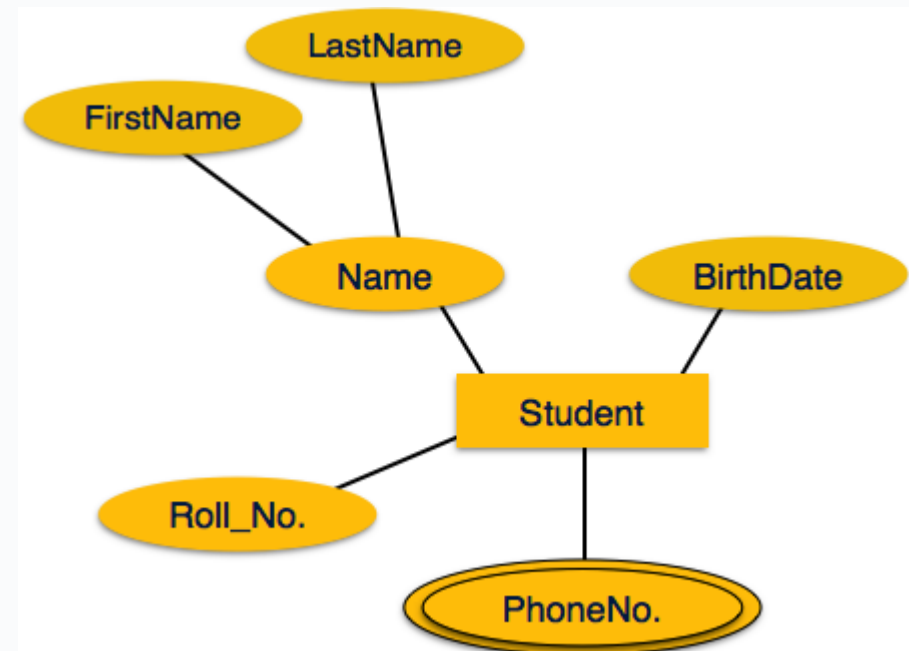
Composite Attributes

- **Composite** attributes are further divided in a tree like structure.
- Every node is then connected to its attribute. That is, composite attributes are represented by ellipses that are connected with an ellipse.



Multivalued Attributes

- **Multivalued** attributes are depicted by double ellipse.



Entity-Relationship Model: Keys

- **Identifier/key:**

- A minimal set of attributes for an entity set, such that each entity in the set can be uniquely identified.
- In some cases, there may be a single attribute (such as PPS#) which serves as a key, but in some models you might need multiple attributes as a key (*'Bob from Accounting'*).
- There may be several possible candidate keys. You will generally designate one such key as the **primary key**.

Entity-Relationship Model: Keys

- **Characteristics of Identifiers:**
 - Will not change in value.
 - Will not be null.
 - No intelligent identifiers (e.g. containing locations or people that might change).

Entity-Relationship Model

- **Relationship** – The logical association among entities is called *relationship*.
- Relationships are mapped with entities in various ways.
- Mapping cardinalities define the number of association between two entities.

Tips on building entity-relationship diagrams

- ***Nouns*** are clues to the entities of a business
 - Start by describing the work that is done in the project or department or area of interest, then
 - pull out nouns in the sentences as potential candidates for entities. e.g.

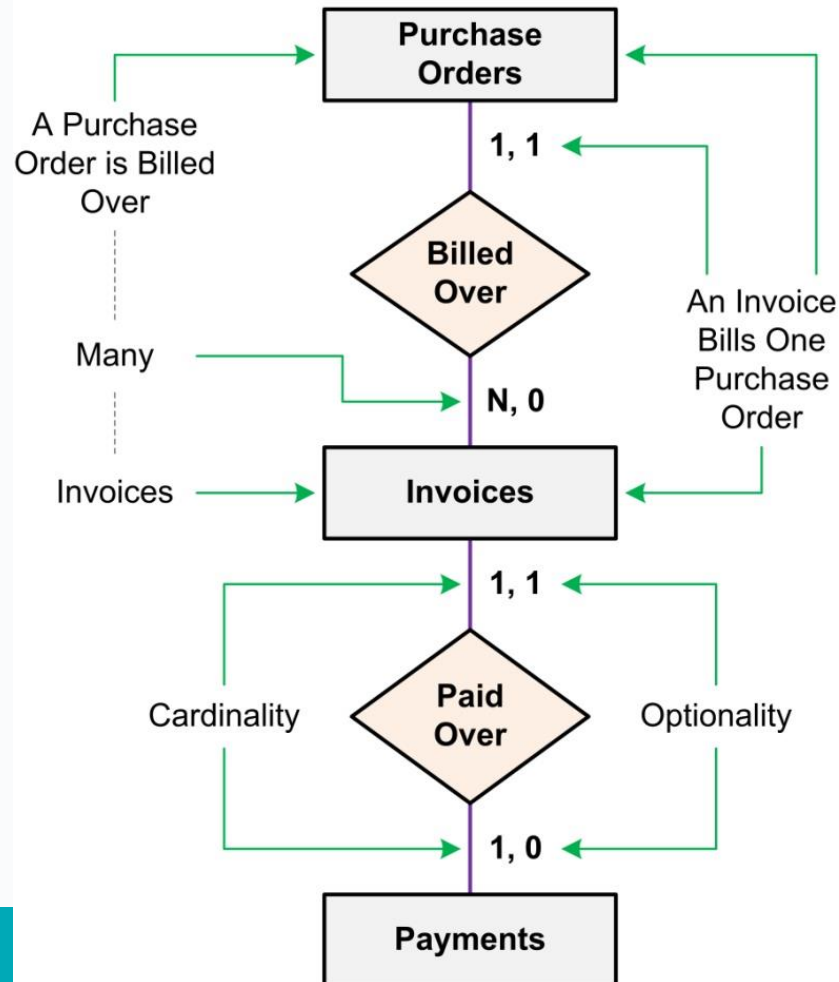
“My address book contains addresses and telephone numbers for both friends and businesses.”

Tips on building entity-relationship diagrams

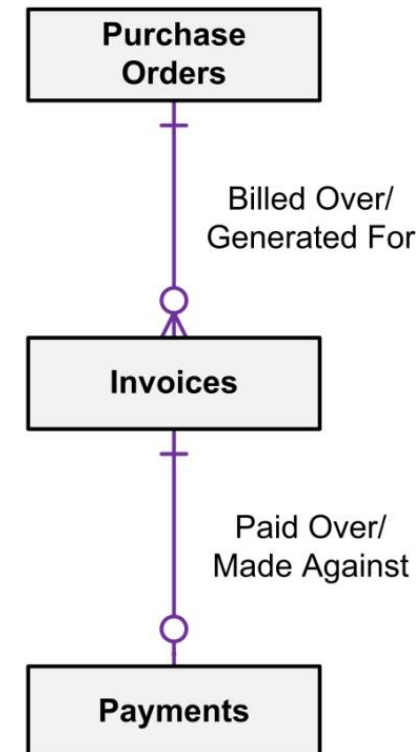
- **Nouns** are clues to the entities of a business
- **Verbs** and **adjectives** sometimes help to get at relationships between entities e.g. '*friends have addresses*'

E-R Diagram Notation

Peter Chen Notation



Bachman Notation

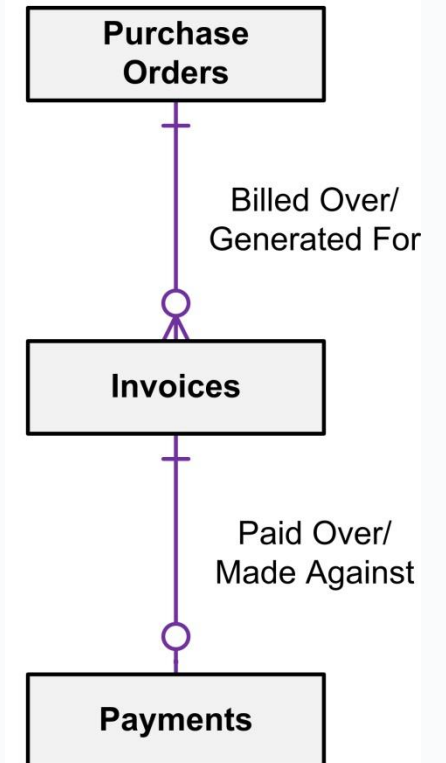


Crow's feet

E-R Diagram Notation

- **Entities** are drawn in boxes.
 - entities should be expressed in the plural.
- The existence of a ***relationship*** between two entities is drawn as a line connecting the entities.

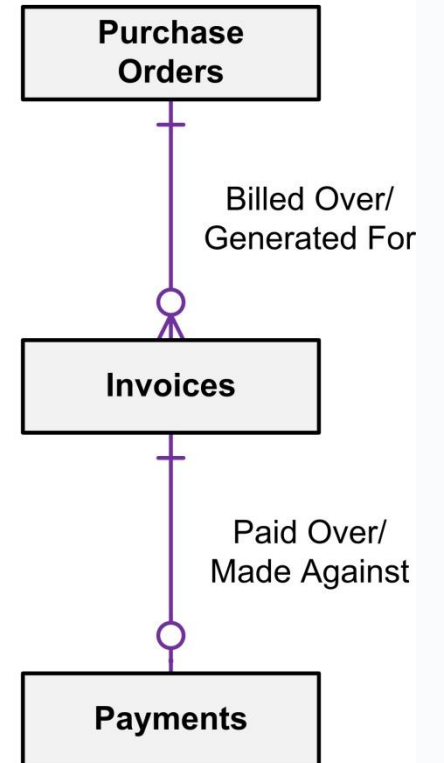
Bachman Notation



E-R Diagram Notation

- **Verbs** may be placed on the relationship lines that describe what the relationships mean e.g.
 - ‘*Purchase Orders are billed over Invoices*’, where
 - ‘*Purchase Orders*’ and ‘*Invoices*’ are related **entities**, and
 - ‘*billed*’ is the **verb** describing the relationship

Bachman Notation



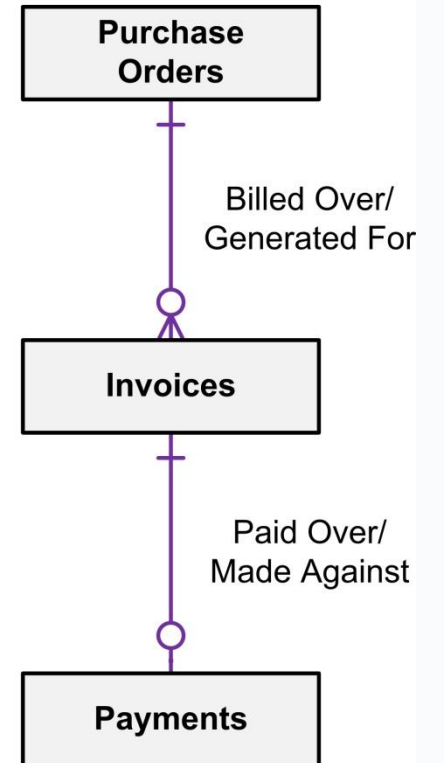
E-R Diagrams: Attributes

- Attributes can be shown on ER models:
 - They appear inside ovals and are attached to their entity.
 - Note that entity types can have a large number of attributes. If all are shown then the diagrams would be confusing. Only show an attribute if it adds information to the ER diagram, or clarifies a point.

E-R Diagram Notation

- Lists of *attributes* can be developed as you go.
 - An *attribute* is a single piece of information that is descriptive of an entity, e.g. friends name, friends e-mail.

Bachman Notation



Exercise

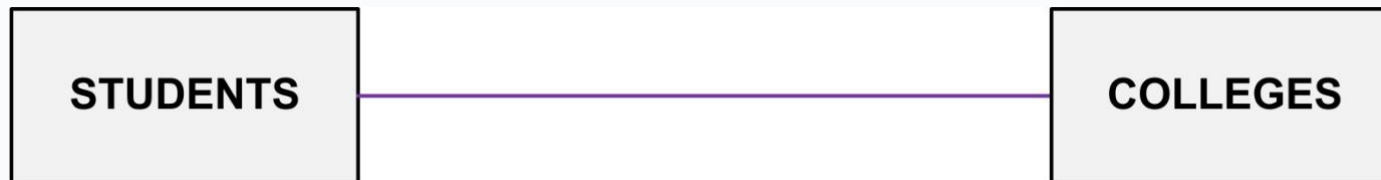
- A restaurant owner hires you to create a database for their business. They need to keep track of customer details (for marketing purposes). A good management of orders, food items, menus, payments and different types of staff is necessary in order to obtain a database that will allow the owner to control all aspects of their business.
- Can you discuss the requirements and come up with a list of entities and attributes that will be required for the DB?

Write your final lists. One member of the group will present them to the whole class.

Let's share your ideas!

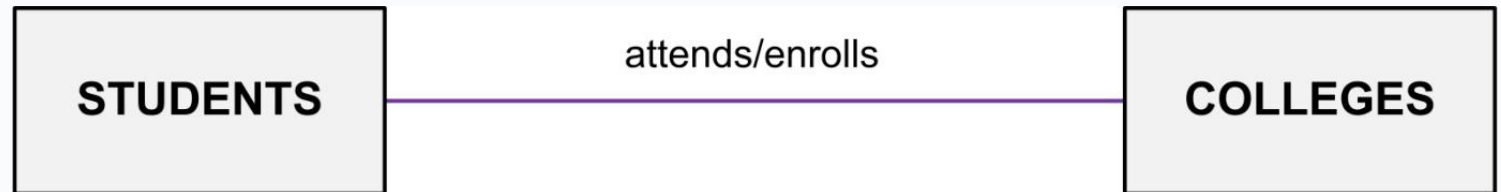
E-R Diagrams: Relationships

- A **relationship** is an association among two or more entities.
 - *The relationship must be uniquely identified by the participating entities.*
- Relationships are represented by lines between entities.



E-R Diagrams: Relationships

- A relationship can also have **descriptive attributes**, to record additional information about the relationship (as opposed to about any one participating entity).



In this case, one can infer that a student may attend a college, or that a college may enroll students.

E-R Diagrams: Optionality and cardinality

- Symbols at the ends of the relationship lines indicate the *optionality* and the *cardinality* of each relationship.
 - *Optionality* expresses whether the relationship is optional or mandatory.
 - *Cardinality* expresses the maximum number of relationships.

E-R Diagrams: Optionality and cardinality

- **Optionality** is of two types...
 1. **Mandatory** Relationship: Associated with every instance of the first entity there will be at least one instance of the second entity.
 2. **Optional** Relationship: There may be instances of the first entity, which are not associated with any instance of the second entity.
 - For example, employee-spouse relationship has to be optional because there could be unmarried employees. It is not correct to make the relationship mandatory.

E-R Diagrams: Optionality and cardinality

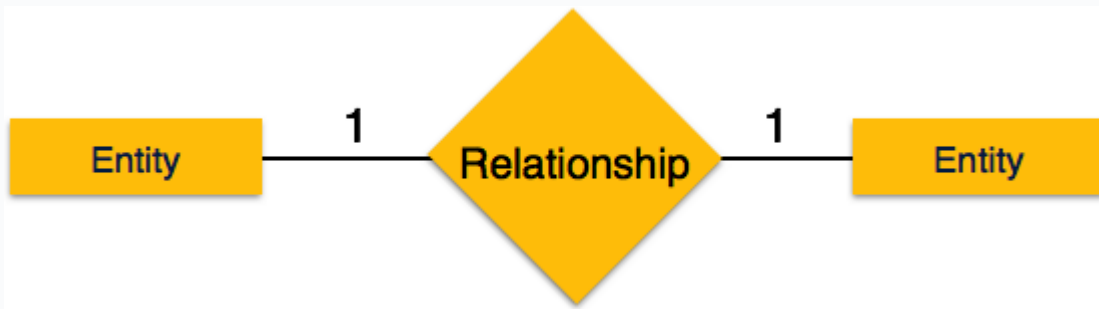
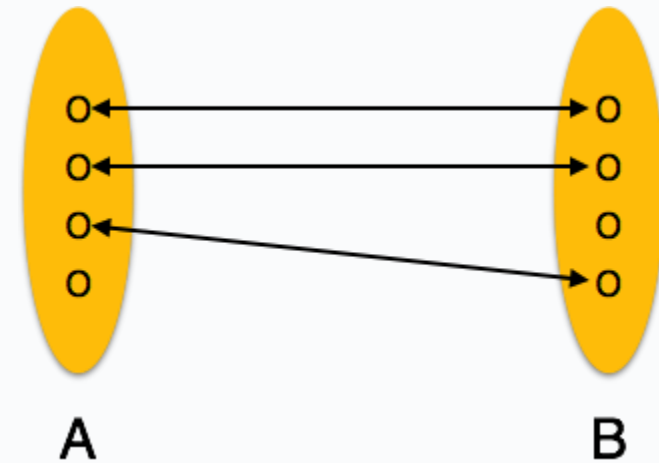
- **Cardinality** is of three types...

One-to-one Relationship:

- An instance of the first entity is associated with only one instance of the second entity. Similarly, each instance of the second entity is related to one instance of the first entity.

One to One

- One entity from entity set A can be associated with at most one entity of entity set B



E-R Diagrams: Optionality and cardinality

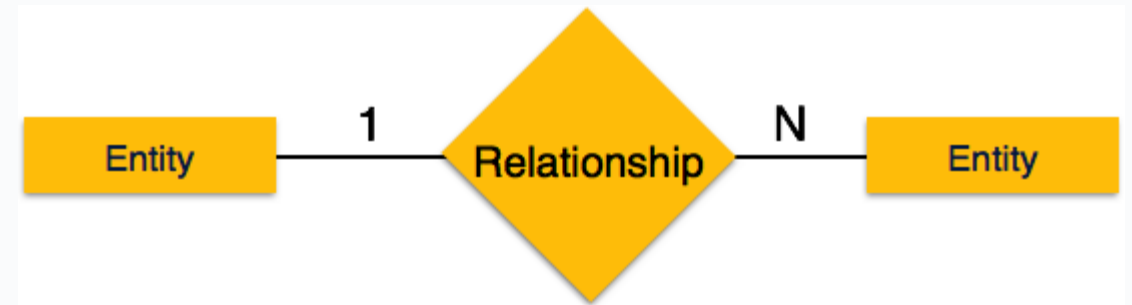
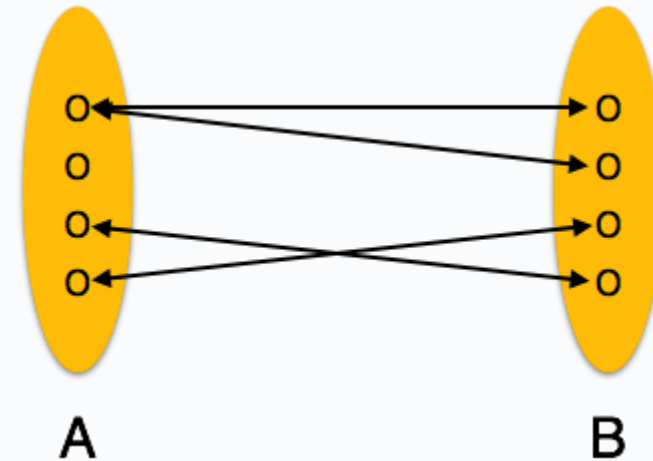
- **Cardinality** is of three types...

One-to-many Relationship:

- One instance of the first entity is related to many instances of the second entity, while an instance of the second entity is associated with only one instance of the first entity.

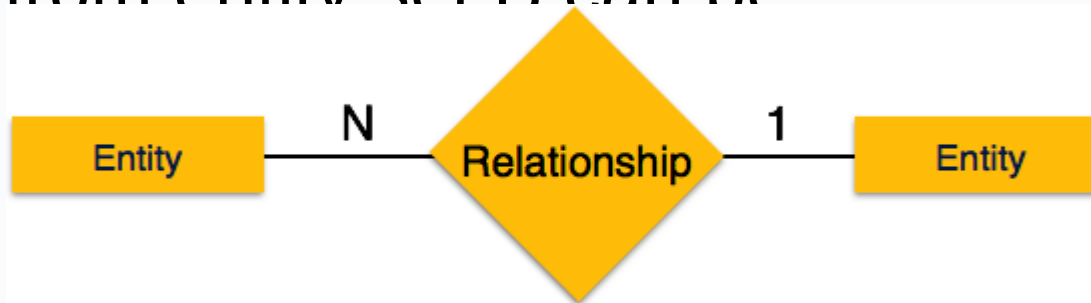
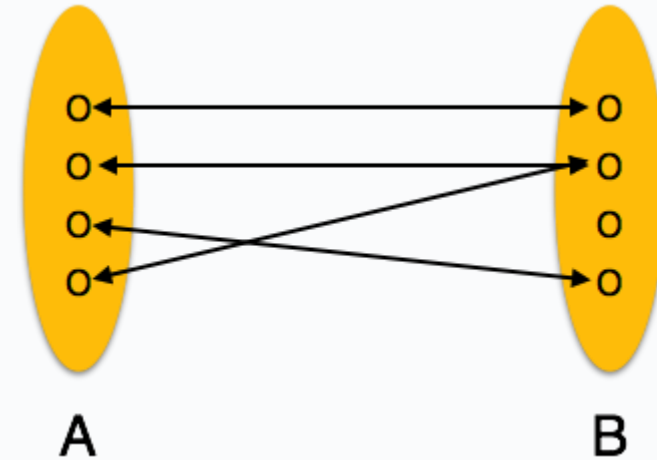
One to Many

- One entity from entity set A can be associated with more than one entities of entity set B however an entity from entity set B, can be associated with at most one entity.



Many to One

- More than one entity from entity set A can be associated with at most one entity of entity set B, however an entity from entity set B can be



E-R Diagrams: Optionality and cardinality

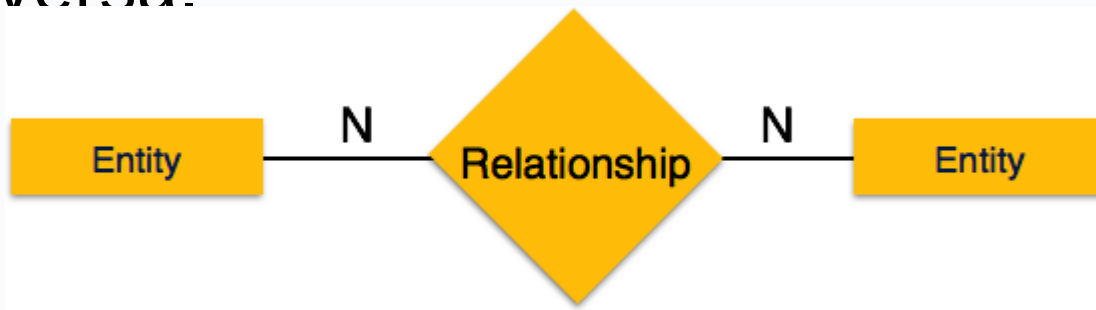
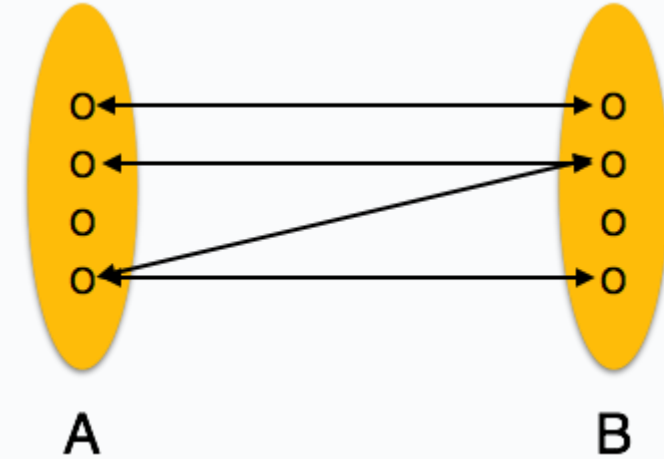
- **Cardinality** is of three types...

Many-to-many Relationship:

- An instance of the first entity is related to many instances of the second entity and the same is true in the reverse direction also.

Many to Many

- One entity from A can be associated with more than one entity from B and vice versa.

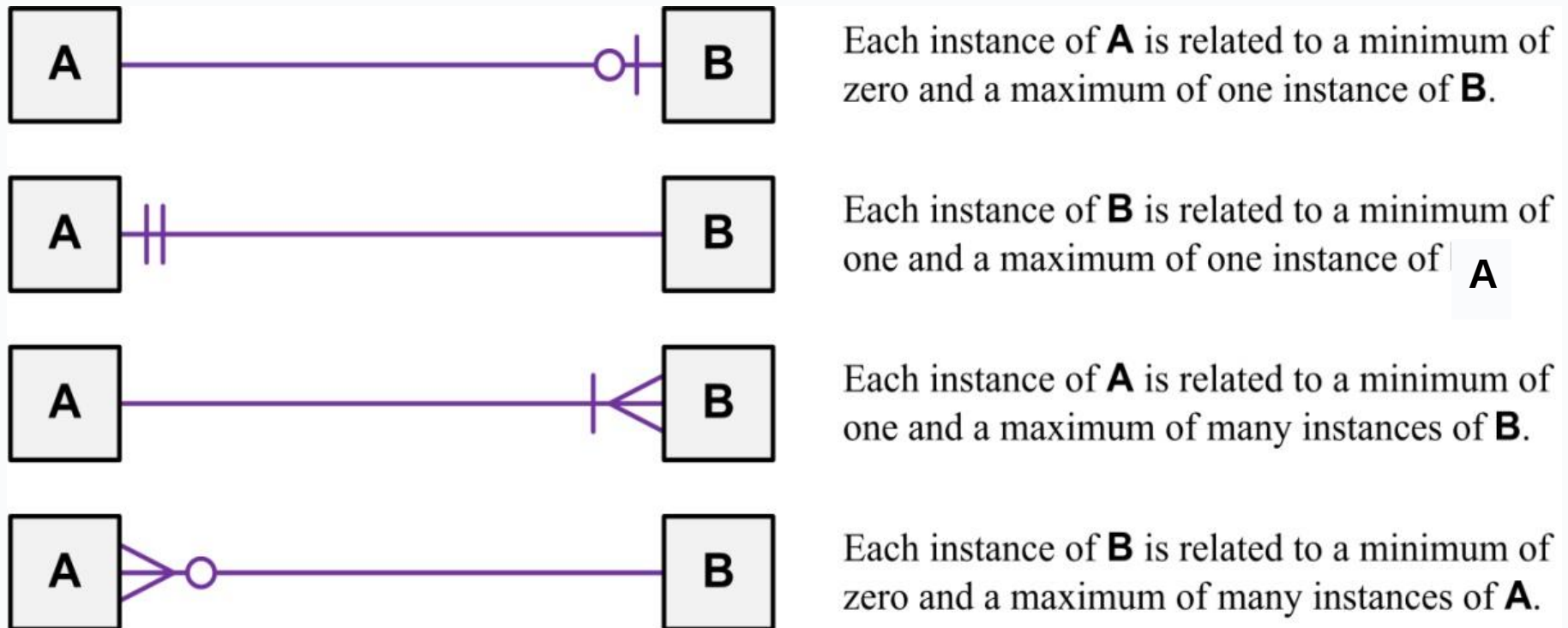


E-R Diagrams: Optionality and cardinality

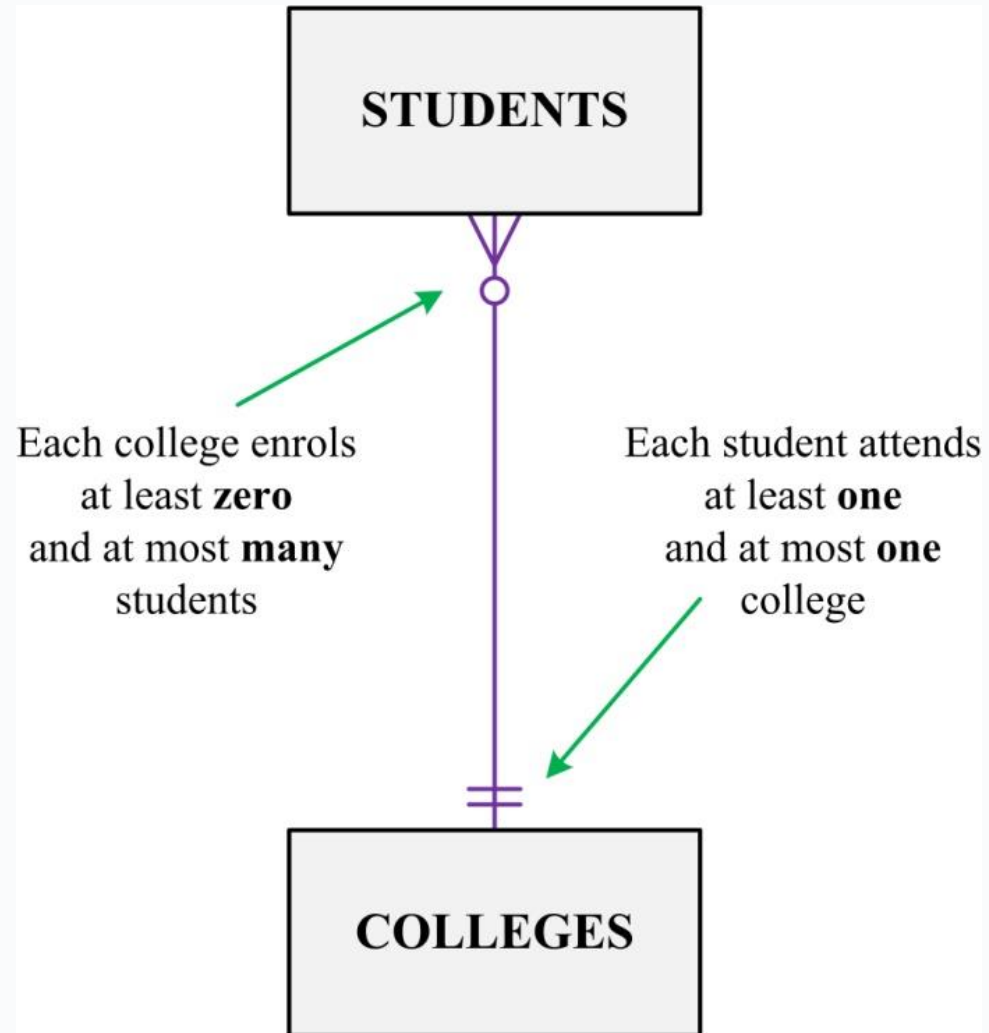
Symbol	Meaning
	One and only one i.e. mandatory one
	One or more i.e. mandatory many
	Zero or one i.e. optional one
	Zero or more i.e. optional many

Cardinality E-R Notation

E-R Diagrams: Optionality and cardinality



E-R Diagrams: Optionality and cardinality



Tips

- **DO NOT** start diagramming first!
 - Write down the entities.
 - Write down the attributes for each entity.
 - Select the identifying attribute(s) for each entity.

Tips

- Draw a rough draft of the E-R diagram. Show entities and relationships only.
 - Add cardinalities.
 - Add attributes.

The ERD Modelling Process

- **Initially...**
 - Identify obvious ***entities*** (i.e. Person, Place, Object, Event, Concept)
 - Identify obvious ***relationships***
 - Identify properties of ***entities***
 - Underline Primary key property
- **Refine the Model**
 - The initial ***entities*** and ***relationships*** may not be sufficient

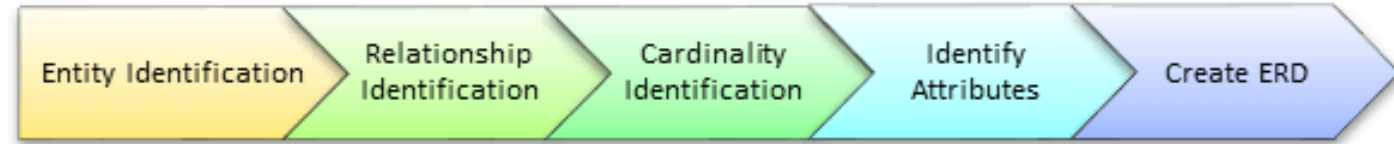
The ERD Modelling Process

- Then...
 - Decide on ***Relationship Constraints***
 - ***Cardinality*** Ratios
 - ***Optionality*** constraints
 - Check for completeness
 - Has all the required information been represented

Summary: Constructing an E-R model

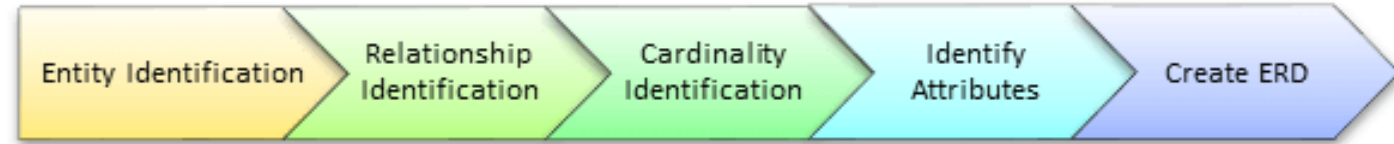
1. Identify Entities
2. List the Initial Attributes of Each Entity
3. Mark the Primary Keys
4. Define the Relationships
5. Describe the Cardinality & Optionality of the Relationships
6. Complete list of attributes

Example



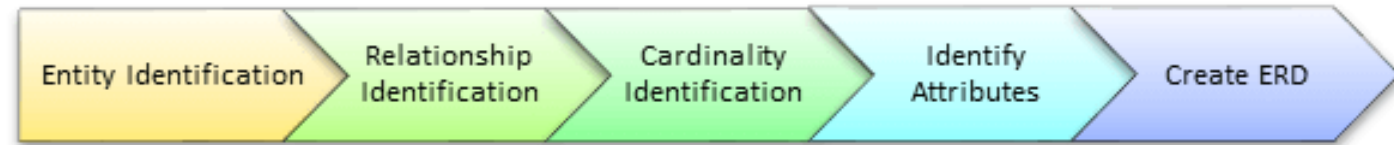
- In a university, a Student enrolls in Modules.
- A student must be assigned to at least one or more Modules. Each module is taught by a single lecturer.
- To maintain instruction quality, a lecturer can deliver only one module.

Example



- **Step 1) Entity Identification**

Example

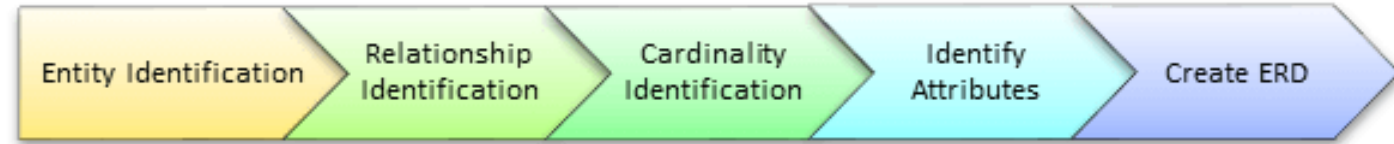


- **Step 1) Entity Identification**

We have three entities

- Student
- Module
- Professor

Example

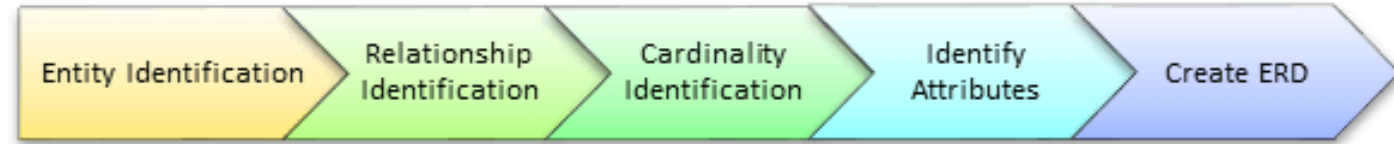


- **Step 2) Relationship Identification**

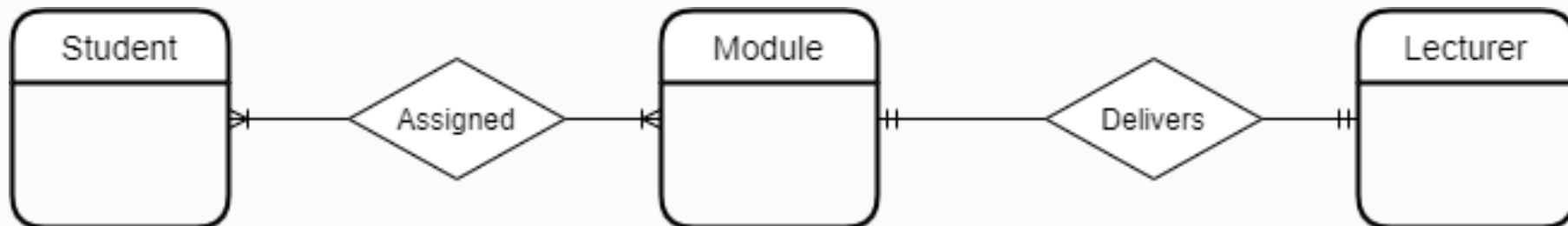
We have the following two relationships

- The student is **assigned** a module
- Lecturer **delivers** a module

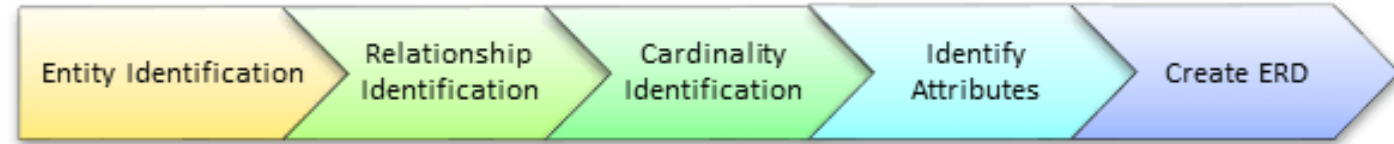
Example



- **Step 3) Cardinality Identification**
- From the problem statement we know that,
 - A student can be assigned **multiple** modules
 - A lecturer can deliver only **one** module

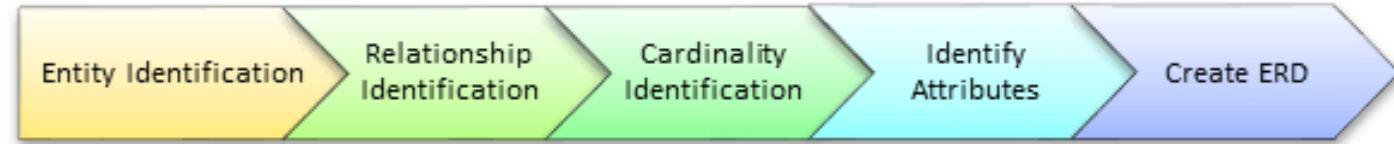


Example



- **Step 4) Identify Attributes**

Example



- **Step 4) Identify Attributes**

Name

Semester

Surname

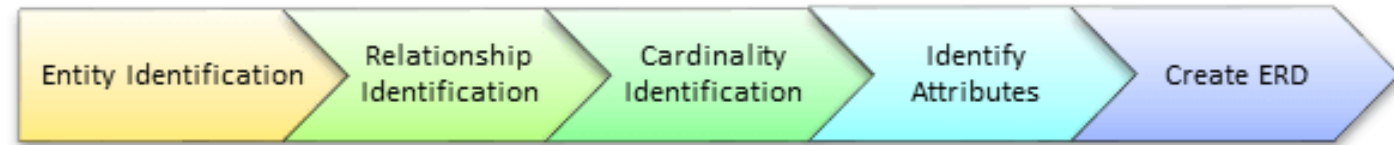
Credits

Student ID

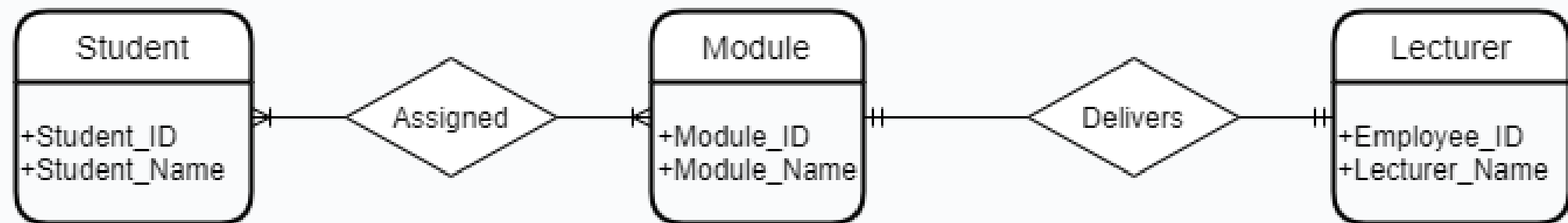
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Course ID

Example



- **Revise**



Thanks and reference

- Edmund Nevin for earlier versions of these lecture notes
- [https://www.tutorialspoint.com/dbms/er_diagram_representation
.htm](https://www.tutorialspoint.com/dbms/er_diagram_representation.htm)



**Break
Back at 8pm
for Lab session**

**TU257
Information Systems
Marisa Llorens**